Lecture 1
Systems of Linear Equations
- Matrices

1.1 Introduction to Systems of Linear Equations

1.2 Gaussian Elimination and Gauss-Jordan Elimination

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1.1 Introduction to Systems of Linear Equations

A linear equation in *n* variables:

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b\ ,$$

$a_1, a_2, a_3, \ldots, a_n, b$: real numbers – a_i coefficient x_i variable or unknown

a_1: leading coefficient

x_1: leading variable or unknown

- Notes:

(1) Linear equations have no products or roots of variables and no variables involved in trigonometric, exponential, or logarithmic functions.

(2) Variables appear only to the first power.

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Ex 1: (Linear or Nonlinear)

Linear $(a)\ 3x+2y=7$

$(b)\ \frac{1}{2}x+y-\pi z=\sqrt{2}$ Linear

Linear $(c)\ x_1-2x_2+10x_3+x_4=0$

$(d)\ (\sin\frac{\pi}{2})x_1-4x_2=e^2$ Linear

Exponential

Nonlinear $(e)\ xy+z=2$

$(f)\ e^x-2y=4$ Nonlinear

Product of variables

Nonlinear $(g)\ \sin x_1+2x_2-3x_3=0$

$(h)\ \frac{1}{x}+\frac{1}{y}=4$ Nonlinear

Trigonometric function

not the first power

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- a solution of a linear equation in *n* variables

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b$$

is a set of *n* real numbers

$$x_1 = s_1,\ x_2 = s_2,\ x_3 = s_3,\ \ldots,\ x_n = s_n$$

such that

$$a_1s_1 + a_2s_2 + a_3s_3 + \cdots + a_ns_n = b$$

Solution set:

the set of all solutions of a linear equation

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Ex 2: (Parametric representation of a solution set)

$$x_1 + 2x_2 = 4$$

a solution: (2, 1), i.e. $x_1 = 2, x_2 = 1$

another solution: (4,0), i.e. $x_1=4$, $x_2=0$

If you solve for x_1 in terms of x_2, you obtain

$$x_1 = 4 - 2x_2,$$

By letting $x_2 = t$ you can represent the solution set as

$$x_1 = 4 - 2t$$

And the solutions are $\{(4-2t, t) \mid t \in R\}$ or $\{(s, 2-\frac{1}{2}s) \mid s \in R\}$

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A system of *m* linear equations in *n* variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_m \end{cases}$$

To solve the system of linear equations is to find **all** set of *n* real numbers $x_1 = s_1, x_2 = s_2, x_3 = s_3, \cdots, x_n = s_n$ satisfying all equations of the system at the same time – called *solution*.

- Consistent:

  A system of linear equations has *at least one solution*.

- Inconsistent:

  A system of linear equations has *no solution*.

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Notes:

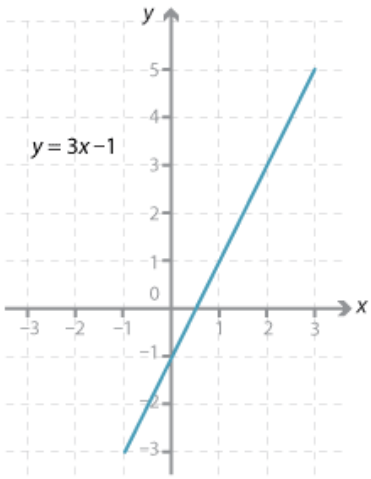
Every system of linear equations has either

(1) exactly one solution,

(2) infinitely many solutions, or

(3) no solution.

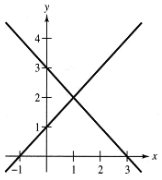
Example: a linear equation of two variables determines a straight line in the plane



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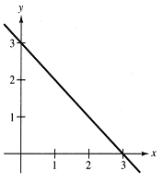
Ex 4: (Solution of a system of linear equations)

(1) $x + y = 3$
$x - y = -1$
two intersecting lines



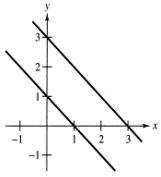
exactly one solution

(2) $x + y = 3$
$2x + 2y = 6$
two coincident lines



Infinitely many solutions

(3) $x + y = 3$
$x + y = 1$
two parallel lines



no solution

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Ex 5: (Using back substitution to solve a system in row echelon form)

$$\begin{aligned} x - 2y &= 5 \quad (1) \\ y &= -2 \quad (2) \end{aligned}$$

Sol: By substituting $y = -2$ into (1), you obtain

$$\begin{aligned} x - 2(-2) &= 5 \\ x &= 1 \end{aligned}$$

The system has exactly one solution: $x = 1, y = -2$

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Ex 6: (Using back substitution to solve a system in row echelon form)

$$\begin{aligned} x - 2y + 3z &= 9 \quad (1) \\ y + 3z &= 5 \quad (2) \\ z &= 2 \quad (3) \end{aligned}$$

Sol: Substitute $z = 2$ into (2)

$$\begin{aligned} y + 3(2) &= 5 \\ y &= -1 \end{aligned}$$

and substitute $y = -1$ and $z = 2$ into (1)

$$\begin{aligned} x - 2(-1) + 3(2) &= 9 \\ x &= 1 \end{aligned}$$

The system has exactly one solution:

$x = 1, y = -1, z = 2$

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Equivalent:

Two systems of linear equations are called **equivalent** if they have precisely the same solution set.

- Notes:

  Eliminations: use following operations on a system of linear equations to produce an equivalent system.

  (1) Interchange two equations.

  (2) Multiply an equation by a nonzero constant.

  (3) Add a multiple of an equation to another equation.

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Ex 7: Solve a system of linear equations (consistent system)

$$\begin{aligned} x - 2y + 3z &= 9 \quad (1)\\ -x + 3y &= -4 \quad (2)\\ 2x - 5y + 5z &= 17 \quad (3) \end{aligned}$$

Sol: $(1) + (2) \rightarrow (4)$

$$\begin{aligned} x - 2y + 3z &= 9\\ y + 3z &= 5 \quad (4)\\ 2x - 5y + 5z &= 17 \end{aligned}$$

$(1)\times(-2) + (3) \rightarrow (5)$

$$\begin{aligned} x - 2y + 3z &= 9\\ y + 3z &= 5\\ -y - z &= -1 \quad (5) \end{aligned}$$

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$(4)+(5)\rightarrow(6)$

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2z &= 4 \end{aligned} \qquad (6)$$

$(6)\times\frac{1}{2}\rightarrow(6)$

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ z &= 2 \end{aligned}$$

So the solution is $x=1, y=-1, z=2$ (only one solution)

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Ex 8: Solve a system of linear equations (inconsistent system)

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 1 && (1) \\ 2x_1 - x_2 - 2x_3 &= 2 && (2) \\ x_1 + 2x_2 - 3x_3 &= -1 && (3) \end{aligned}$$

Sol:

$(1)\times(-2)+(2)\rightarrow(4)$

$(1)\times(-1)+(3)\rightarrow(5)$

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 1 \\ 5x_2 - 4x_3 &= 0 \qquad (4) \\ 5x_2 - 4x_3 &= -2 \qquad (5) \end{aligned}$$

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$(4)\times(-1)+(5)\to(5)$

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 1 \\ 5x_2 - 4x_3 &= 0 \\ 0 &= -2 \end{aligned}$$ (a false statement)

So, the system has no solution (an inconsistent system).

Ex 9: Solve a system of linear equations (infinitely many solutions)

$$\begin{aligned} x_2 - x_3 &= 0 && (1) \\ x_1 - 3x_3 &= -1 && (2) \\ -x_1 + 3x_2 &= 1 && (3) \end{aligned}$$

Sol: $(1)\leftrightarrow(2)$

$$\begin{aligned} x_1 - 3x_3 &= -1 && (1) \\ x_2 - x_3 &= 0 && (2) \\ -x_1 + 3x_2 &= 1 && (3) \end{aligned}$$

$(1)+(3)\to(4)$

$$\begin{aligned} x_1 - 3x_3 &= -1 \\ x_2 - x_3 &= 0 \\ 3x_2 - 3x_3 &= 0 \end{aligned} \qquad (4)$$

$$\begin{aligned} x_1 \quad &- \quad 3x_3 \quad = \quad -1 \\ x_2 \quad &- \quad x_3 \quad = \quad 0 \end{aligned}$$

$\Rightarrow x_2 = x_3, \quad x_1 = -1 + 3x_3$

Let $x_3 = t$ any real number

then $x_1 = 3t - 1,$

$x_2 = t, \qquad t \in R$

$x_3 = t,$

So this system has infinitely many solutions.

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1.2 Gaussian Elimination and Gauss-Jordan Elimination

$m \times n$ matrix: (m *by* n matrix)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

m rows

n columns

- Notes:

(1) Every entry a_{ij} in a matrix is a number.

(2) A matrix with m rows and n columns is said to be of size $m \times n$.

(3) If $m = n$, then the matrix is called square of order n.

(4) For a square matrix, the entries a_{11}, a_{22}, …, a_{nn} are called the main diagonal entries.

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Ex 1:

Matrix	Size
$[2]$	1×1
$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	2×2
$\begin{bmatrix} 1 & -3 & 0 & \frac{1}{2} \end{bmatrix}$	1×4
$\begin{bmatrix} e & \pi \\ 2 & \sqrt{2} \\ -7 & 4 \end{bmatrix}$	3×2

- Note:

  One very common use of matrices is to represent a system of linear equations.

A system of m linear equations in n variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_m \end{cases}$$

Matrix form: $Ax = b$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ & \vdots & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- Coefficient matrix:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ & \vdots & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = A$$

- Augmented matrix:

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} & b_3 \\ & \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} & b_m \end{array}\right] = [A \mid b]$$

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Elementary row operation:

(1) Interchange two rows. $r_{ij} : R_i \leftrightarrow R_j$

(2) Multiply a row by a nonzero constant. $r_i^{(k)} : (k)R_i \rightarrow R_i$

(3) Add a multiple of a row to another row. $r_{ij}^{(k)} : (k)R_i + R_j \rightarrow R_j$

- Row equivalent:

Two matrices are said to be **row equivalent** if one can be obtained from the other by a finite sequence of **elementary row operation**.

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Ex 2: (Elementary row operation)

$$\begin{bmatrix} 0 & 1 & 3 & 4 \\ -1 & 2 & 0 & 3 \\ 2 & -3 & 4 & 1 \end{bmatrix} \xrightarrow{r_{12}} \begin{bmatrix} -1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 4 \\ 2 & -3 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & 6 & -2 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix} \xrightarrow{r_1^{(\frac{1}{2})}} \begin{bmatrix} 1 & -2 & 3 & -1 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 2 & 1 & 5 & -2 \end{bmatrix} \xrightarrow{r_{13}^{(-2)}} \begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 0 & -3 & 13 & -8 \end{bmatrix}$$

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Ex 3: Using elementary row operations to solve a system

Linear System	Associated Augemented Matrix	Elementary Row Operation
$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$	$\begin{bmatrix} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{bmatrix}$	
$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2x - 5y + 5z &= 17 \end{aligned}$	$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 2 & -5 & 5 & 17 \end{bmatrix}$	$r_{12}^{(1)} : (1)R_1 + R_2 \to R_2$
$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ -y - z &= -1 \end{aligned}$	$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{bmatrix}$	$r_{13}^{(-2)} : (-2)R_1 + R_3 \to R_3$

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Linear System	Associated Augemented Matrix	Elementary Row Operation
$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2z &= 4 \end{aligned}$	$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix}$	$r_{23}^{(1)} : (1)R_2 + R_3 \to R_3$
$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ z &= 2 \end{aligned}$	$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$	$r_3^{(\frac{1}{2})} : (\frac{1}{2})R_3 \to R_3$
$\longrightarrow \quad \begin{aligned} x &= 1 \\ y &= -1 \\ z &= 2 \end{aligned}$		

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A matrix is of row-echelon form if it satisfies (1), (2), (3)

A matrix is of reduced row-echelon form if it satisfies (1), (2), (3) and (4)

(1) All row consisting entirely of zeros occur at the bottom of the matrix.

(2) For each row that does not consist entirely of zeros, the first nonzero entry is 1 (called **a leading 1**).

(3) For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row.

(4) Every column that has a leading 1 has zeros in every position above and below its leading 1.

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Ex 4: (Row-echelon form or reduced row-echelon form)

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$ (row - echelon form)

$$\begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ (reduced row - echelon form)

$$\begin{bmatrix} 1 & -5 & 2 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$ (row - echelon form)

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ (reduced row - echelon form)

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & -4 \end{bmatrix}$$

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Gaussian elimination:

The procedure for reducing a matrix to a row-echelon form.

Gauss-Jordan elimination:

The procedure for reducing a matrix to a reduced row-echelon form.

- Notes:

(1) Every matrix has an unique reduced row echelon form.

(2) A row-echelon form of a given matrix is not unique.

(Different sequences of row operations can produce different row-echelon forms.)

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Ex: (Procedure of Gaussian elimination and Gauss-Jordan elimination)

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 8 & 12 \\ 2 & 8 & -6 & 4 & 12 & 28 \\ 2 & 8 & -1 & 4 & -5 & 4 \end{bmatrix} \xrightarrow{r_{12}} \begin{bmatrix} 2 & 8 & -6 & 4 & 12 & 28 \\ 0 & 0 & -2 & 0 & 8 & 12 \\ 2 & 8 & -1 & 4 & -5 & 4 \end{bmatrix}$$

Produce leading 1

The first nonzero column

$$\xrightarrow{r_1^{(\frac{1}{2})}} \begin{bmatrix} 1 & 4 & -3 & 2 & 6 & 14 \\ 0 & 0 & -2 & 0 & 8 & 12 \\ 2 & 8 & -1 & 4 & -5 & 4 \end{bmatrix} \xrightarrow{r_{13}^{(-2)}} \begin{bmatrix} 1 & 4 & -3 & 2 & 6 & 14 \\ 0 & 0 & -2 & 0 & 8 & 12 \\ 0 & 0 & 5 & 0 & -17 & -24 \end{bmatrix}$$

leading 1

Zeros elements below leading 1

Produce leading 1

The first nonzero column

Submatrix

$$\xrightarrow{r_2^{(-\frac{1}{2})}} \begin{bmatrix} 1 & 4 & -3 & 2 & 6 & 14 \\ 0 & 0 & 1 & 0 & -4 & -6 \\ 0 & 0 & 5 & 0 & -17 & -24 \end{bmatrix} \xrightarrow{r_{23}^{(-5)}} \begin{bmatrix} 1 & 4 & -3 & 2 & 6 & 14 \\ 0 & 0 & 1 & 0 & -4 & -6 \\ 0 & 0 & 0 & 0 & 3 & 6 \end{bmatrix}$$

leading 1

Zeros elements below leading 1

Submatrix

Produce leading 1

Zeros elsewhere

$$\xrightarrow{r_3^{(\frac{1}{3})}} \begin{bmatrix} 1 & 4 & -3 & 2 & 6 & 14 \\ 0 & 0 & 1 & 0 & -4 & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_{31}^{(-6)}} \begin{bmatrix} 1 & 4 & -3 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 & -4 & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

leading 1

(row - echelon form)

(row - echelon form)

$$\xrightarrow{r_{32}^{(4)}} \begin{bmatrix} 1 & 4 & -3 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_{21}^{(3)}} \begin{bmatrix} 1 & 4 & 0 & 2 & 0 & 8 \\ 0 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

(row - echelon form)

(reduced row - echelon form)

Ex 7: Solve a system by Gauss-Jordan elimination method (only one solution)

$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

Sol:

augmented matrix

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{bmatrix} \xrightarrow{r_{12}^{(1)},\, r_{13}^{(-2)}} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{bmatrix} \xrightarrow{r_{23}^{(1)}} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\xrightarrow{r_3^{(\frac{1}{2})}} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_{21}^{(2)},\, r_{32}^{(-3)},\, r_{31}^{(-9)}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \longrightarrow \begin{aligned} x &= 1 \\ y &= -1 \\ z &= 2 \end{aligned}$$

(row - echelon form) (reduced row - echelon form)

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Ex 8 : Solve a system by Gauss-Jordan elimination method (infinitely many solutions)

$$\begin{cases} 2x_1 + 4x_2 - 2x_3 = 0 \\ 3x_1 + 5x_2 = 1 \end{cases}$$

Sol: augmented matrix

$$\begin{bmatrix} 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{bmatrix} \xrightarrow{r_1^{(\frac{1}{2})},\, r_{12}^{(-3)},\, r_2^{(-1)},\, r_{21}^{(-2)}} \begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$ (reduced row - echelon form)

the corresponding system of equations is

$$\begin{aligned} x_1 + 5x_3 &= 2 \\ x_2 - 3x_3 &= -1 \end{aligned}$$

leading variable : x_1, x_2

free variable : x_3

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$$x_1 = 2 - 5x_3$$
$$x_2 = -1 + 3x_3$$

Let $x_3 = t$

$$x_1 = 2 - 5t,$$
$$x_2 = -1 + 3t, \qquad t \in R$$
$$x_3 = t,$$

So this system has infinitely many solutions.

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Homogeneous systems of linear equations:

A system of linear equations is said to be homogeneous if all the constant terms are zero.

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0\\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0\\
a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= 0\\
&\vdots\\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= 0
\end{aligned}$$

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Trivial solution:

$$x_1 = x_2 = x_3 = \cdots = x_n = 0$$

- Nontrivial solution:

  other solutions

- Notes:

(1) Every homogeneous system of linear equations is consistent.

(2) If the homogenous system has fewer equations than variables, then it must have an infinite number of solutions.

(3) For a homogeneous system, exactly one of the following is true.

(a) The system has only the trivial solution.

(b) The system has infinitely many nontrivial solutions in addition to the trivial solution.

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Ex 9: Solve the following homogeneous system

$$\begin{aligned} x_1 - x_2 + 3x_3 &= 0 \\ 2x_1 + x_2 + 3x_3 &= 0 \end{aligned}$$

Sol: augmented matrix

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix} \xrightarrow{r_{12}^{(-2)}, r_2^{(\frac{1}{3})}, r_{21}^{(1)}} \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$ (reduced row-echelon form)

leading variable : x_1, x_2

free variable : x_3

Let $x_3 = t$

$x_1 = -2t, x_2 = t, x_3 = t, t \in R$

When $t = 0, x_1 = x_2 = x_3 = 0$ (trivial solution)

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